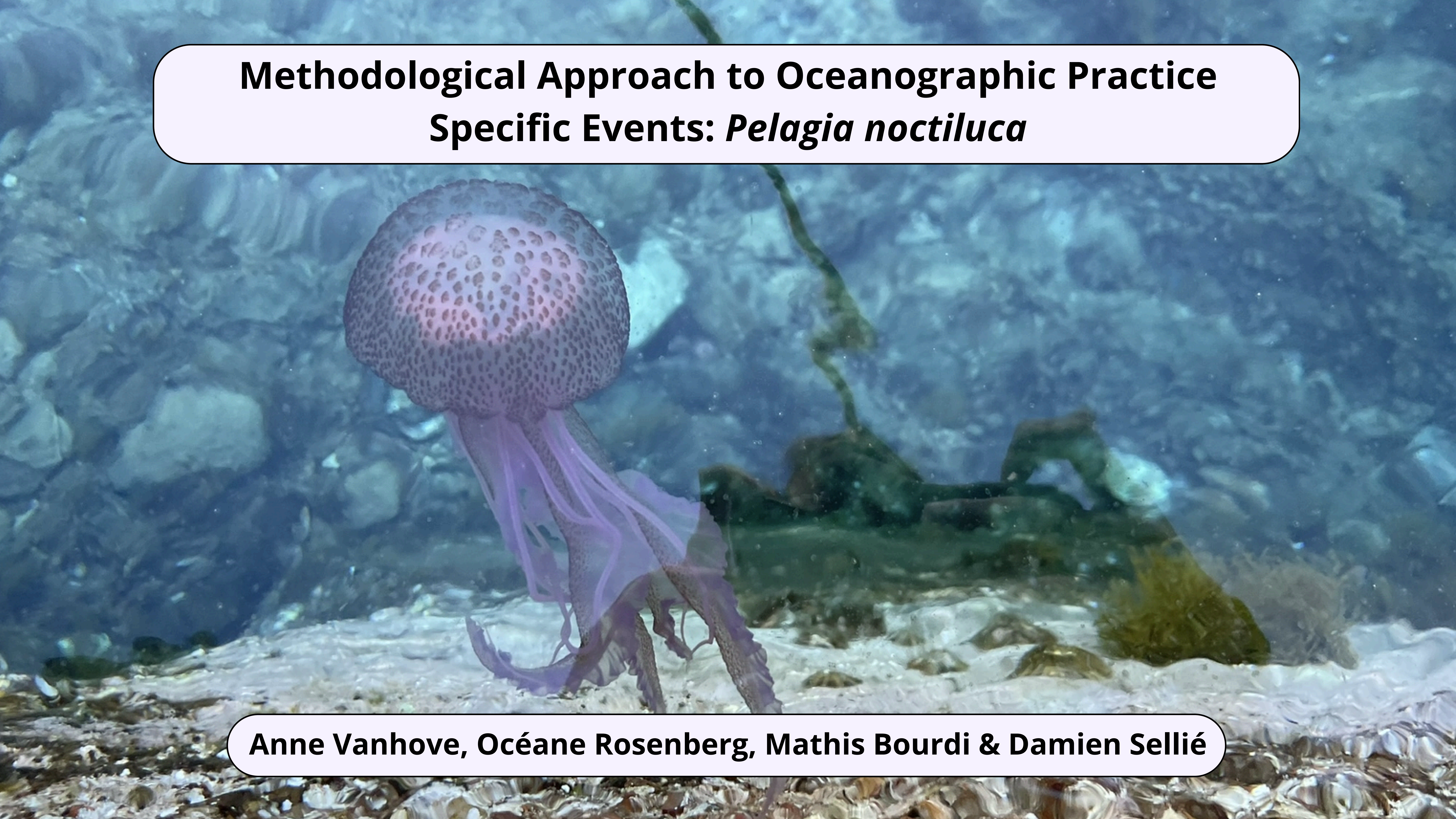
# Methodological Approach to Oceanographic Practice Specific Events: *Pelagia noctiluca*



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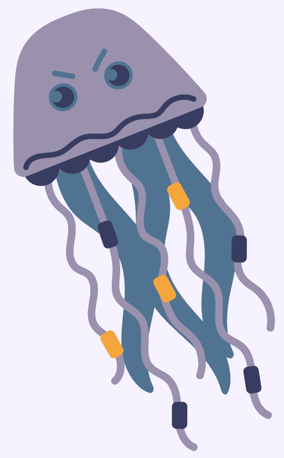
# Introduction

**<u>Caracteristics of the Mediterranean Sea</u>**

- Oligotrophic sea; Average salinity : 38; Average temperature : 21°C
    - Surface water warming → proliferation of certain species (=*Pelagia noctiluca)*



***<u>Pelagia noctiluca</u>***

- Abundant and venomous
- Negative impact on human activities
- Mainly offshore distribution
- Diel vertical migration influenced by the day/night cycle and zooplankton movements

**<u>Hydrodynamics conditions</u>**

- Transport of jellyfish to coastal areas influenced by specific currents and weather conditions (wind)

# Parameters influencing the abundance of *Pelagia noctiluca* in Calvi Bay

- **Circulation in Calvi Bay**

→ complex and influenced by the Liguro-Provençal current and winds

- **Factors influencing Pelagia noctiluca blooms**
  - Water temperature (life cycle linked to seasonal variations, higher temperatures favor reproduction and development)
  - Plankton availability
  - Winds

- **Interactions between factors**
  - Upwelling : increases primary production and enhances food availability for zooplankton and jellyfish
  - Winds: modify surface water temperature by mixing water layers, thereby influencing jellyfish living conditions

# Description of *Pelagia noctiluca*

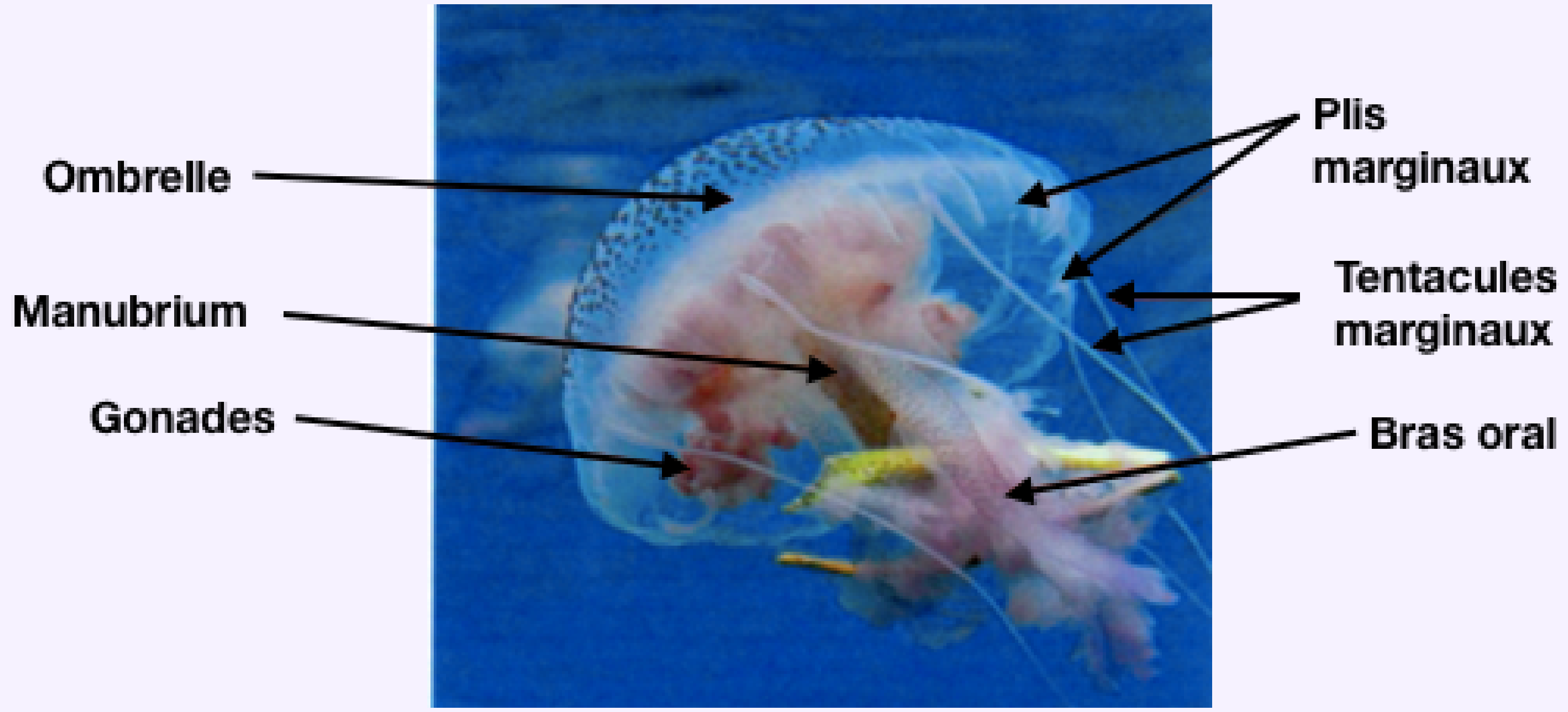



**Figure 1:** External anatomy of *Pelagia noctiluca* (Hecq et al., 2009)

**Specimens collected in the harbor, measured and weighed:**

| **Individuals** (n=10) | **Length (cm)** | **Width (cm)** | **Weight (g)** |
|---|---|---|---|
| **Mean** | 12,2 | 6,27 | 56,7 |

- Body size ranges from 3 to 15 cm (adults), covered with stinging cnidocytes
- Tentacles and oral arms capture and paralyze prey (zooplankton, crustaceans, jellyfish, small fish)

# Counting method

Objective of the study : To investigate the relationship between environmental parameters and jellyfish abundance.

Counting method : Surface counts of jellyfish conducted along the harbor over a fixed distance
Standardization : 2m-wide transect (low glare and minimal reflection)

Counts permormed twice a day : Morning (7h30) and evening (18h30)
⟶ Calculation of daily abundance

Development of an abundance scale for Sylvain's data, ranging from 0 to 3 :

0 = No jellyfish
1 = < 10 jellyfish
2 = 10 to 50 jellyfish
3 = > 50 jellyfish

# Transect area calculation

**1:** Measurement of the research area perimeter
**2:** Observations conducted over a 2-meter width
**3:** Area calculation performed using QGIS

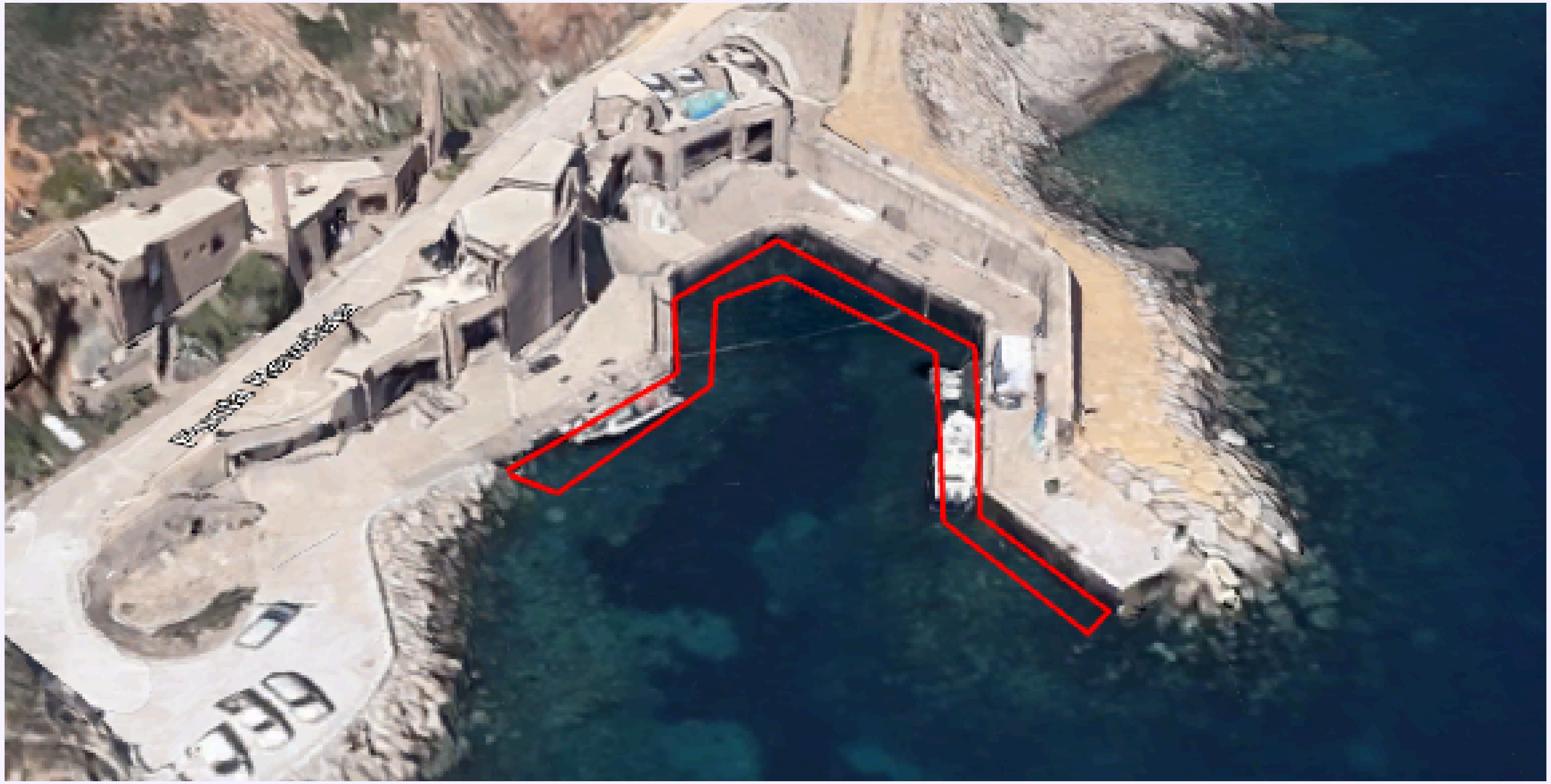


**Figure 2:** *Pelagia noctiluca* counting area mapped using QGIS

# Small scale – Harbor

**Figure 3:** Photo of the STARESO harbor (ULiège)

Weather mainly sunny (9 out of 12 days)

Air temperature ranged between 16°C and 26°C

Average water temperature: 16,7°C

Light winds with Beaufort scale 0–1 on 8 out of 12 days

- <u>Wind direction :</u>
- Undirected winf (4j)
- Northeast wind (3j)
- Northwest wind (2j)
- South wind (2j)
- North wind (1j)

Jellyfish abundance varied from **0** up to a total of **164** individuals counted along the transect

# Statistical analysis (R)

A generalized linear model was used to assess the effects of wind, temperature, and seasonal variations on daily jellyfish abundance

## <u>Summary of significant results :</u>

**Wind strength** significantly negative = stronger wind reduce surface jellyfish abudnance (inverse relationship)

**Northeast wind:** significant ($p < 0.1$) → NE winds appear to increase jellyfish abundance

**Southeast wind**: significantly negative ($p < 0.05$) → SE winds reduce jellyfish abundance

**Spring season:** significant ($p < 0.05$) → jellyfish abundance is higher in spring

# Medium scale – GITAN weather station

**<u>Winds :</u>**

Data from weather
**Station GITAN**

Highlighted days = presence of jellyfish

Correlation between
- North, northwest and northeast winds
- jellyfish presence

| Date | Wind direction (morning) | Wind direction (noon) | Wind direction (evening) |
|---|---|---|---|
| 05/05/2024 | Northwest | Northeast | Northwest |
| 06/05/2024 | Northwest | Northwest | North |
| 07/05/2024 | Northwest | Northeast | North |
| 08/05/2024 | West | West | Northwest |
| 09/05/2024 | West | Southwest | Southeast |
| 10/05/2024 | West | Southwest | West |
| 11/05/2024 | Northwest | North | Northwest |
| 12/05/2024 | Northwest | North | Northwest |
| 13/05/2024 | Northwest | Northeast | Northeast |
| 14/05/2024 | West | Southwest | Northeast |
| 15/05/2024 | North | East | Northwest |
| 16/05/2024 | Northwest | East | East |

**Table 1:** Summary table of wind directions in the morning, at noon, and in the evening from May 5 to 16, 2024

# Medium scale – GITAN weather station

**Water temperature :**

Daily mean water temperature of the harbor

**ΔT° = 1.5°C**

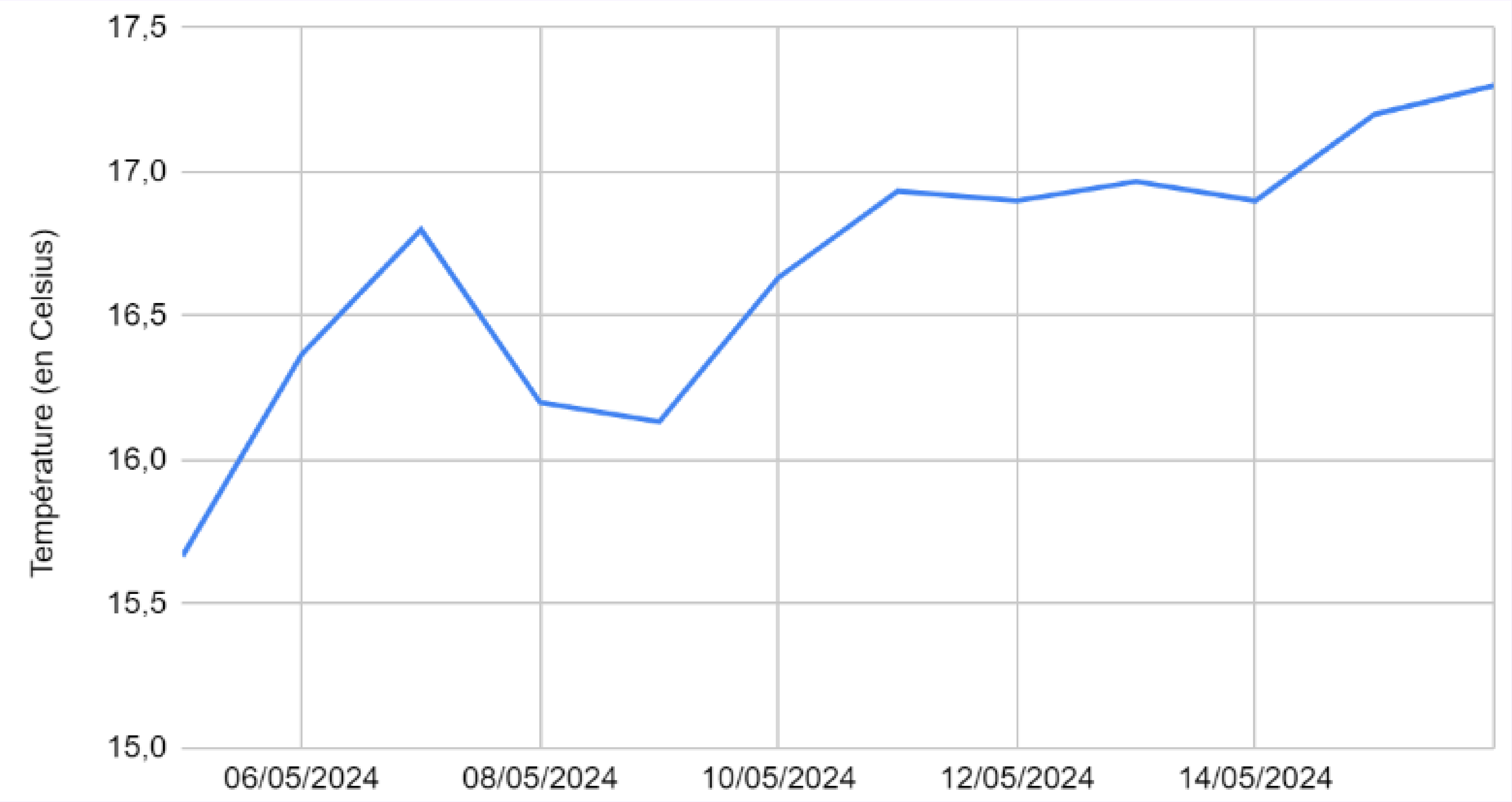



**Figure 4:** Graph of mean surface temperature in Stareso harbor between May 5 and May 16

# Medium scale – GITAN weather station

**La température de l'eau :**

Daily mean water temperature of the harbor

**ΔT° = 1.5°C**

Framed periods = jellyfish presence

Weak correlation between temperature increase and jellyfish presence

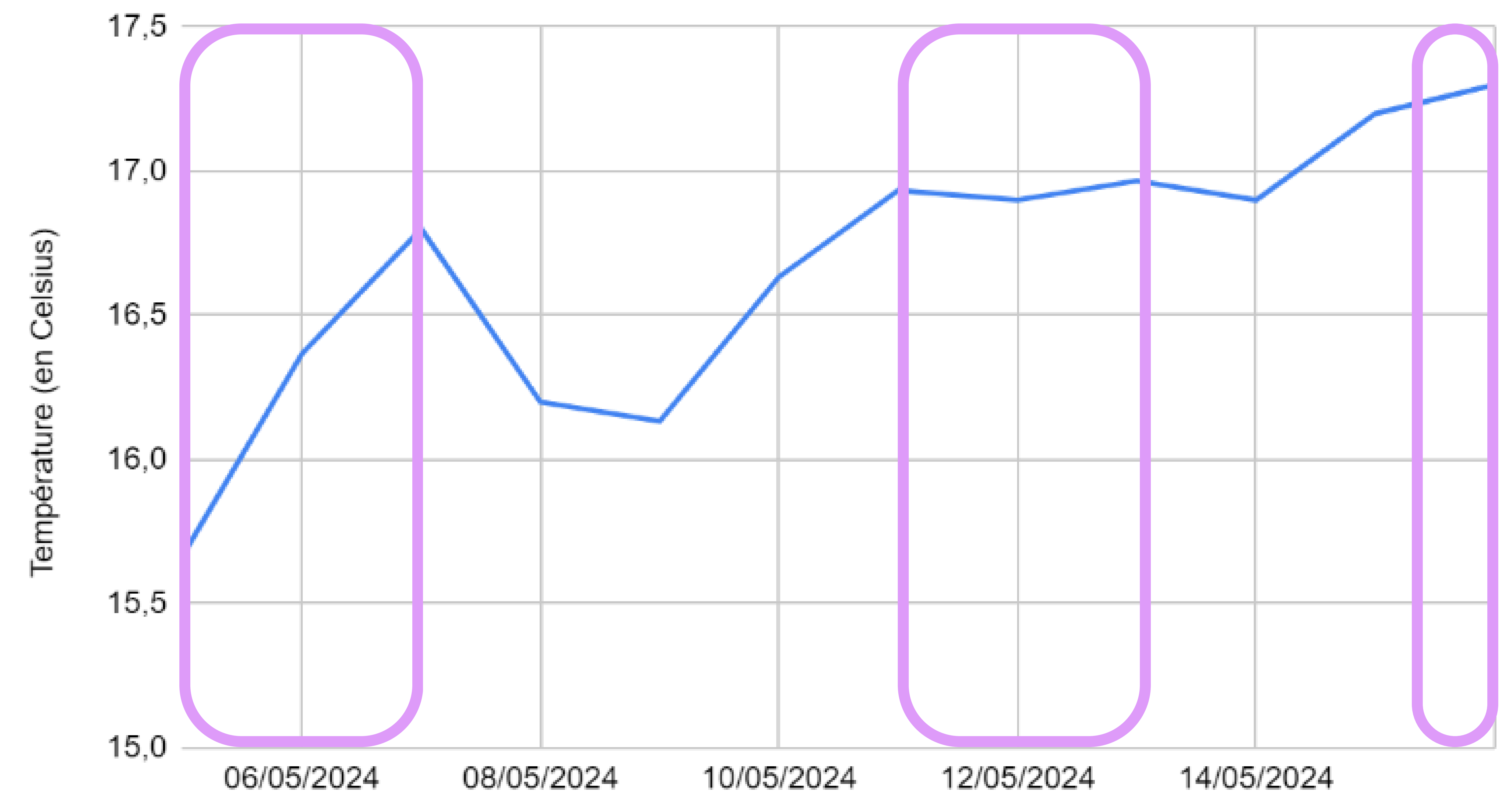
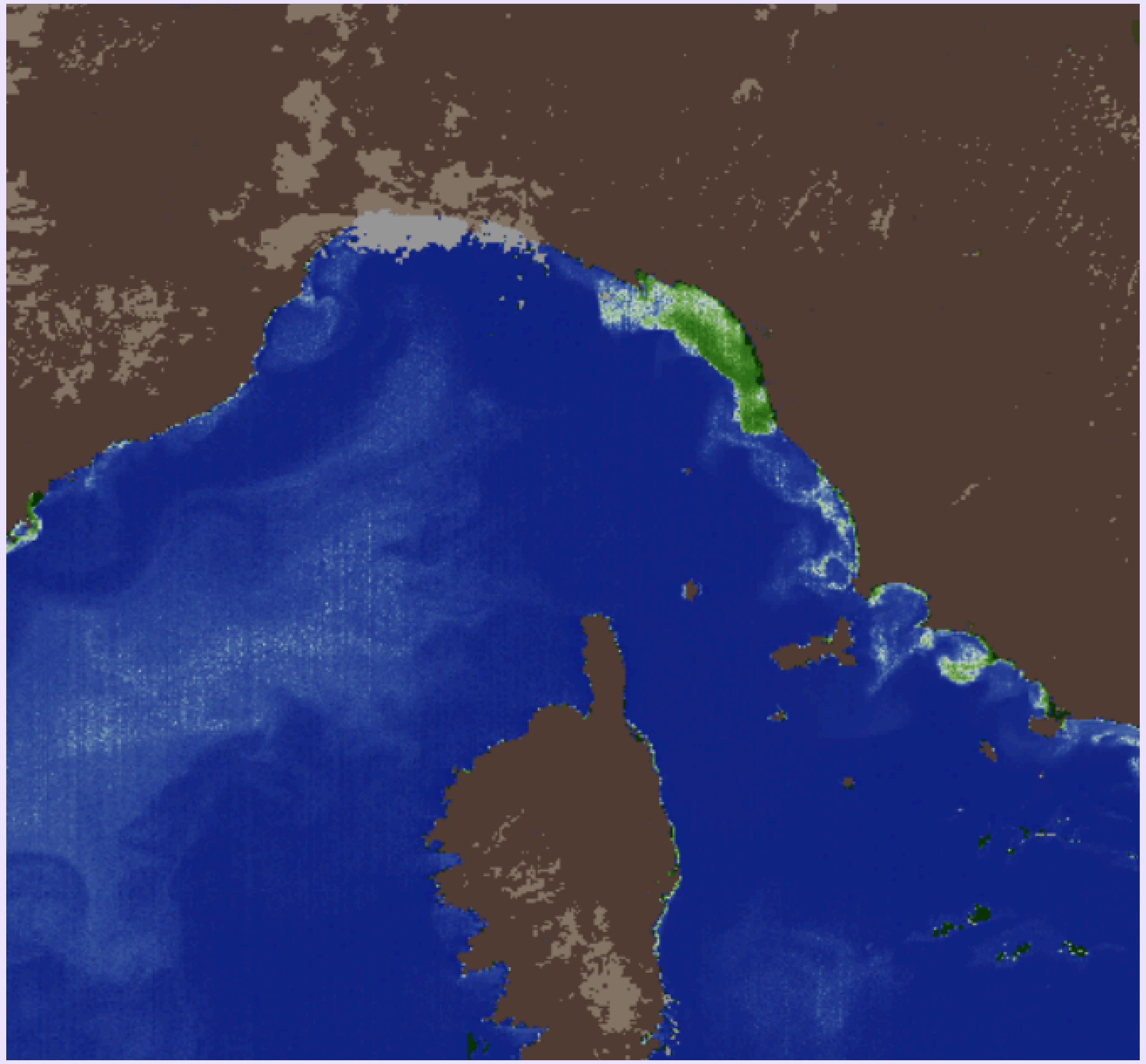


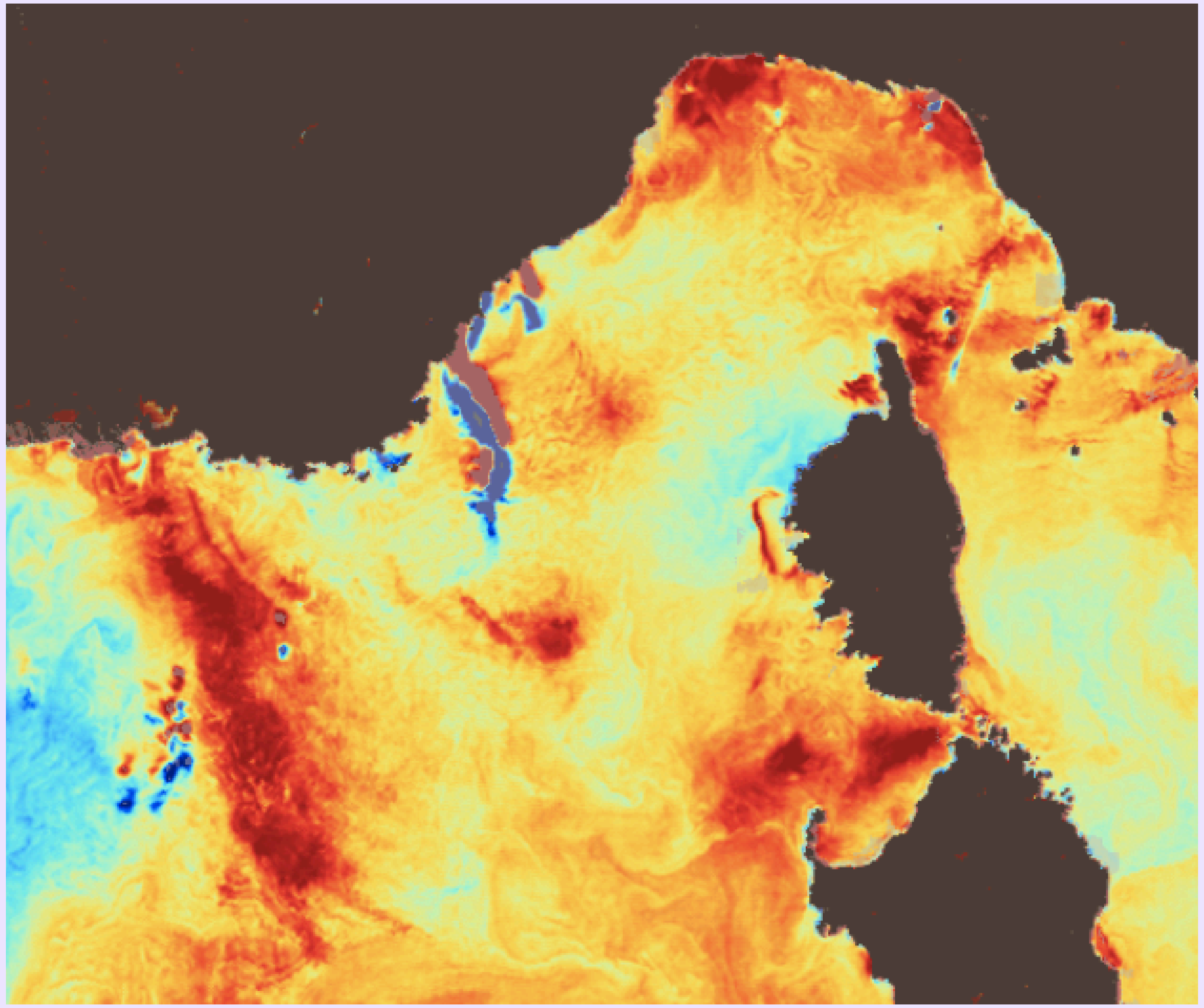

**Figure 4:** Graph of mean surface temperature in Stareso harbor from May 5 to May 16

# Large scale – Satellite data

**Chl a** (May 10, 2024)



**SST** (May 10, 2024)



**Figures 5 and 6:** Maps of Chla and SST in the Liguro-Provençal basin obtained from SNAP for May 10, 2024

→ Not correlated with jellyfish presence

→ Favorable for jellyfish development

## Discussion

**Environmental factors influencing Pelagia noctiluca abundance:**

- North winds favor high jellyfish abundance
- Spring is correlated with an increase in jellyfish abundance
- Temperature shows no observable short-term effect

- **Diet of *Pelagia noctiluca***
    - Mainly copepods and cladocerans (90%)
    - Remaining 10%  = chaetognaths
    - Low copepods abundance may influence jellyfish abundance

## Discussion

- **Protocole de comptage**
  - Simple mais généraliste : pas forcément adapté à l'espèce
  - Méduses en surface pour minimiser les biais dûs à la visibilité et aux conditions de lumière
  - Limitations dues à l'influence des vents et à la forme en entonnoir du port
  - Les méduses étaient agglutinées au fond du port devant le laboratoire
  - Pas d'estimation fiable de l'abondance totale

- **Amélioration suggérées pour l'étude**
  - Etendre la durée de l'étude pour identifier des tendances saisonnières plus claires et comprendre les variations interannuelles
  - Renforcer les tests statistiques pour réduire la sensibilité aux valeurs aberrantes
  - Analyser d'autres facteurs environnementaux (circulation des eaux de surface, salinité, disponibilité de nourriture)

# Thank you for your attention

## Any questions ?